
DISCRETE STRUCTURES I

Chapter : Mathematical Induction

Departement of Computer Science – IT College – University of Bahrain

Dr. Amine Mahjoub, Dr. Mohamed Mazin – 2022/2023 – Semester 1

1 Introduction

Mathematical induction is an extremely important proof technique that can be used to prove results about a large variety of discrete objects. For example, it is used to prove results about the complexity of algorithms, the correctness of certain types of computer programs, theorems about graphs and trees, as well as a wide range of identities and inequalities. In this chapter, we will describe how mathematical induction can be used and why it is a valid proof technique.

2 Principle

In general, mathematical induction can be used to prove statements that assert that $P(n)$ is true for all positive integers n , where $P(n)$ is a propositional function. A proof by mathematical induction has two parts, a basis step, where we show that $P(1)$ is true, and an inductive step, where we show that for all positive integers k , if $P(k)$ is true, then $P(k+1)$ is true.

PRINCIPLE OF MATHEMATICAL INDUCTION : To prove that $P(n)$ is true for all positive integers n , where $P(n)$ is a propositional function, we complete two steps :

1. BASIS STEP : We verify that $P(1)$ is true.
2. INDUCTIVE STEP : We show that the conditional statement $P(k) \longrightarrow P(k+1)$ is true for all positive integers k .

To complete the inductive step of a proof using the principle of mathematical induction, we assume that $P(k)$ is true for an arbitrary positive integer k and show that under this assumption, $P(k+1)$ must also be true. The assumption that $P(k)$ is true is called the inductive hypothesis. Once we complete both steps in a proof by mathematical induction, we have shown that $P(n)$ is true for all positive integers, that is, we have shown that $\forall n P(n)$ is true where the quantification is over the set of positive integers. In the inductive step, we show that $\forall k (P(k) \longrightarrow P(k+1))$ is true, where again, the domain is the set of positive integers.

3 Exercises

Exercise 1 :

Let $P(n)$ the statement that $1^2 + 2^2 + \dots + n^2 = n(n+1)(2n+1)/6$ for the positive integer n .

Question 1. What is the statement $P(1)$?

Question 2. Show that $P(1)$ is true, completing the basis step of the proof.

Question 3. What is the inductive hypothesis?

Question 4. Complete the inductive step.

Exercise 2 :

Let $P(n)$ the statement that $1^3 + 2^3 + \dots + n^3 = n^2(n+1)^2/4$ for the positive integer n .

Question 1. What is the statement $P(1)$?

Question 2. Show that $P(1)$ is true, completing the basis step of the proof.

Question 3. What is the inductive hypothesis?

Question 4. Complete the inductive step.

Exercise 3 :

Prove each of the following summation for any nonnegative integer n :

Question 1. $1 + 3 + 5 + \dots + (2n-1) = n^2$ for $n \geq 3$

Question 2. $\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots + \frac{1}{(2n-1)(2n+1)}$ for $n \geq 1$

Question 3. $2 + 5 + 8 + \dots + (3n-1) = \frac{n(3n+1)}{2}$.

Exercise 4 :

Use mathematical induction to prove divisibility facts for any positive integer n :

Question 1. Prove that 2 divides $n^2 + n$.

Question 2. Prove that 3 divides $n^3 + 2n$.

Question 3. Prove that 6 divides $n^3 - n$.

Question 4. Prove that $7^n - 2^n$ is divisible by 5 for any integer $n \geq 1$.

Exercise 5 :

Let

$$T_n = \begin{cases} 6 & \text{if } n = 0 \\ 4T_{n-1} + 2^n & \text{otherwise} \end{cases}$$

Show that $T_n = 7 \cdot 4^n - 2^n$ for $n \geq 0$.

Exercise 6 :

Question 1. Given $T_0 = 2$ and $T_n = T_{n-1} - 3$ show $T_n = 2 - 3n$ for $n \geq 0$.

Question 2. Given $T_1 = 0$ and $T_n = T_{n-1} + 2n + 10$ show $T_n = n^2 + 11n - 12$ for $n \geq 1$.

Question 3. Given $T_0 = 5$ and $T_n = 3T_{n-1} + n^2 - 7n - 1$ show $T_n = \frac{1}{4}(3^{n+1} - 2n^2 + 8n + 17)$ for $n \geq 0$.